

Simple python program-2

```
import numpy as np

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.neural_network import MLPClassifier

from sklearn.metrics import accuracy_score, classification_report


# Load Multi-Class Dataset (Iris Dataset)

data = load_iris()

X = data.data

y = data.target


# Split the dataset into Training and Testing data

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


# Feature Scaling

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)


# Create Neural Network Model

model = MLPClassifier(
    hidden_layer_sizes=(2, 2), # 2 Hidden Layers, 2 Neurons each
    activation='tanh',        # Tanh Activation Function
    learning_rate_init=0.1,    # Learning Rate
```

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max_iter=1000,
random_state=42
)
# Train the Model
model.fit(X_train, y_train)
# Make Predictions
y_pred = model.predict(X_test)
# Calculate Accuracy
accuracy = accuracy_score(y_test, y_pred)
# Display Results
print("Neural Network Analysis for Multi-Class Data")
print("Learning Rate: 0.1")
print("Activation Function: Tanh")
print("Hidden Layers: 2")
print("Hidden Neurons: 2")
print("Accuracy:", accuracy)
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

```

Output

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Neural Network Analysis for Multi-Class Data
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Learning Rate: 0.1
Activation Function: Tanh
Hidden Layers: 2
Hidden Neurons: 2
-----
Accuracy: 1.0

Classification Report:
      precision    recall  f1-score   support

     0         1.00      1.00      1.00        10
     1         1.00      1.00      1.00         9
     2         1.00      1.00      1.00        11

   accuracy          1.00
  macro avg          1.00
weighted avg          1.00

```